

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 – Digital Electronics
Spring Term 2026

Lab. 8- BJT I_C

Issued: 3/18/2026

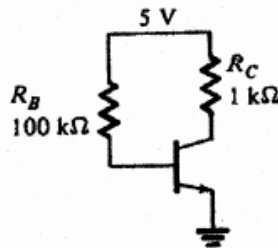
Due: 3/20/2026

Question 1: Based on Problems 5.1 and 5.2 (D.A. Hodges and H.G. Jackson, *Analysis and Design of Digital Integrated Circuits*, 2nd Ed., McGraw-Hill, 1988.)

Objectives:

- BJT forward-active and saturation current calculations

Given the Si NPN BJT below where $\alpha_F = 0.98$, $\alpha_R = 0.49$, $I_{ES} = 10^{-15}\text{A}$:



- a. Solve for I_C and V_{CE} following the steps below:
 - i. Guess that the NPN is in forward-active ($V_{BE} = 0.7\text{V}$, $V_{BC} < 0.5\text{V}$) and calculate I_C .
 - ii. Calculate V_{BE} accurately by reworking $I_C = I_{ES} \cdot e^{V_{BE}/V_T}$.
 - iii. Verify that the NPN is in forward-active (BE junction must be on, BC junction must be off).
 - iv. Calculate V_{CE} .
 - v. Perform a DC operating point simulation in LTSPICE. Report and compare I_C and V_{CE} to hand calculations.
- b. Solve for I_C and V_{CE} following the steps below when $R_B = 10\text{k}\Omega$.
 - i. Guess that the NPN is in saturation ($V_{BE} = 0.7\text{V}$, $V_{BC} = 0.5\text{V}$) and calculate I_C .
 - ii. Calculate V_{BE} accurately by reworking $I_C = I_{ES} \cdot e^{V_{BE}/V_T}$.
 - iii. Verify that the NPN is in saturation ($I_C < \beta I_B$).
 - iv. Calculate V_{CE} .
 - v. Perform a DC operating point simulation in LTSPICE. Report and compare I_C and V_{CE} to hand calculations.
- c. If the NPN Emitter area was increased to 10x its initial area, what would the new I_{ES} value be?
 - i. For this 10x larger NPN to maintain the same I_C , would V_{BE} need to be increased, or decreased?

Report Submission:

Submit your work on LMS under **Week 9/Lab 8**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm

